

Angad Singh Kochhar

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EDUCATION

Purdue University

M.S. Engineering, Robotics / Interdisciplinary Engineering; GPA: 3.84/4.00

West Lafayette, IN

Expected Dec. 2026

Guru Gobind Singh Indraprastha University

B.Tech., Mechanical & Automation Engineering; 84.3%

New Delhi, India

Sep. 2020

ROBOTICS RESEARCH & EXPERIENCE

SCALE Robotics Lab, Purdue University

Student Researcher — Recreating Emergent Tool Use in Multi-Agent Reinforcement Learning

West Lafayette, IN

May 2026 – Present

- Developing configurable physics-based environments in MuJoCo with agents, objects, and interactive tools to study emergent strategies in multi-agent reinforcement learning.
- Building foundations for PPO-based experiments to evaluate agent strategies, interactions, and increasingly complex autonomous behaviors.

Static Systems Electronics Pvt. Ltd.

Assistant Mechanical Engineer

Gurgaon, India

Sep. 2020 – Dec. 2024

- Supported design, fabrication, fit, installation, and commissioning of customized physical-security equipment for government and private-sector clients.
- Served as a field engineer, troubleshooting mechanical and installation issues, guiding site teams, and coordinating directly with customers on equipment operation and safe deployment.

SELECTED PROJECTS

Autonomous Connect-4 with Boston Dynamics Spot

May 2026 – July 2026

- Developed and tested an autonomous Connect-4 system for Boston Dynamics Spot integrating YOLO board perception, AprilTag localization, minimax decision-making, and robotic manipulation.
- Designed manipulation failure detection and automatic recovery; achieved **>90% chip-detection success** and **>85% successful chip placement** across 100+ trials.

Low-Cost Indoor Positioning Using Combined Inertial Navigation and Wireless Positioning

Spring 2026

- Developed the UWB subsystem for a sub-\$100 indoor-positioning prototype; characterized multipath/reflection effects, implemented C++ filtering, and redesigned sensor placement to improve ranging stability.

Gladiator E-Cove: Battle for Logistics

Spring 2025

- Designed and fabricated a dual-flywheel launcher and implemented IR sensing, finite-state-machine logic, ESP32 communication, and motor/RPM control; achieved **85% delivery accuracy** over 100+ shots and the **highest final competition score**.

Spacecraft Autonomy: Multi-Agent Coordination

Aug. 2025 – Nov. 2025

- Built a MATLAB formation-flying simulation using Clohessy–Wiltshire dynamics, incorporated J2 perturbations, and explored PID-based formation maintenance for autonomous multi-spacecraft coordination.

ModifiedNAFNet: Lightweight Image Restoration

Spring 2025

- Reimplemented and modified NAFNet in PyTorch for limited-resource training, achieving **33.66 dB PSNR** and **0.9400 SSIM** on the SIDD dataset with approximately two hours of training.

DED Machine Monitoring Using Acoustic Machine Learning

Spring 2026

- Led a three-person team analyzing acoustic data from a FormAlloy L2 system; developed a Random Forest model and isolated component-level signatures to localize a major anomalous signal to the machine's moving stage.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, Lua

Robotics & Simulation: MuJoCo, ROS 2, CoppeliaSim, RViz, Boston Dynamics Spot, UR5e

Machine Learning: PyTorch, torchvision, scikit-learn, YOLO, Random Forest

Embedded Systems & Hardware: ESP32, Raspberry Pi, UWB, IMU, Arduino, sensor integration

Tools: Git/GitHub, Linux/Ubuntu, VS Code, LaTeX/Overleaf, Arduino IDE, KiCad